

MODULE *messages*

Message constructors for the *Zcash P2P* wire protocol (*ZIP-0204*).

Each operator builds a record with a header (magic number, command name) and a *payload* whose fields mirror the on-wire format. Fields that are irrelevant to the properties being verified are set to constant values to keep the *TLC* state space manageable.

EXTENDS *Naturals*, *FiniteSets*

$\text{MakeVersion}(\text{from}, \text{to}, \text{blocks}) \triangleq [$
 $\quad \text{header} \mapsto [\text{magic} \mapsto 619259748, \text{command} \mapsto \text{"version"}],$
 $\quad \text{payload} \mapsto [$
 $\quad \quad \text{version} \mapsto 170100,$
 $\quad \quad \text{services} \mapsto 0,$
 $\quad \quad \text{addr_recv} \mapsto \text{to},$
 $\quad \quad \text{addr_from} \mapsto \text{from},$
 $\quad \quad \text{nonce} \mapsto 0,$
 $\quad \quad \text{user_agent} \mapsto \text{""},$
 $\quad \quad \text{start_height} \mapsto \text{Cardinality}(\text{blocks}),$
 $\quad \quad \text{relay} \mapsto \text{FALSE}$
 $\quad]$
 $]$

$\text{MakeVerack} \triangleq [$
 $\quad \text{header} \mapsto [\text{magic} \mapsto 619259748, \text{command} \mapsto \text{"verack"}]$
 $]$

$\text{MakePing} \triangleq [$
 $\quad \text{header} \mapsto [\text{magic} \mapsto 619259748, \text{command} \mapsto \text{"ping"}],$
 $\quad \text{payload} \mapsto [\text{nonce} \mapsto 0]$
 $]$

$\text{MakePong} \triangleq [$
 $\quad \text{header} \mapsto [\text{magic} \mapsto 619259748, \text{command} \mapsto \text{"pong"}],$
 $\quad \text{payload} \mapsto [\text{nonce} \mapsto 0]$
 $]$

$\text{MakeInv}(\text{blocks}) \triangleq [$
 $\quad \text{header} \mapsto [\text{magic} \mapsto 619259748, \text{command} \mapsto \text{"inv"}],$
 $\quad \text{payload} \mapsto [$
 $\quad \quad \text{count} \mapsto 1,$
 $\quad \quad \text{inventory} \mapsto \langle [\text{type} \mapsto \text{"MSG_BLOCK"}, \text{hash} \mapsto \text{Cardinality}(\text{blocks})] \rangle$
 $\quad]$
 $]$

$\text{MakeGetHeaders}(\text{blocks}) \triangleq [$
 $\quad \text{header} \mapsto [\text{magic} \mapsto 619259748, \text{command} \mapsto \text{"getheaders"}],$
 $\quad \text{payload} \mapsto [$
 $\quad \quad \text{version} \mapsto 170100,$
 $\quad]$
 $]$

```

    hashCount      ↦ 1,
    blockLocatorHashes ↦  $\langle \text{Cardinality}(\text{blocks}) \rangle$ ,
    hashStop       ↦ 0
  ]
]

MakeHeaders(blocks)  $\triangleq$  [
  header ↦ [magic ↦ 619259748, command ↦ "headers"],
  payload ↦ [
    count ↦ 1,
    headers ↦  $\langle [version \mapsto 170140, prev\_block \mapsto \text{Cardinality}(\text{blocks}), merkle\_root \mapsto 0, bits \mapsto 0, nonce \mapsto 0] \rangle$ 
  ]
]

MakeGetData(blocks)  $\triangleq$  [
  header ↦ [magic ↦ 619259748, command ↦ "getdata"],
  payload ↦ [
    count ↦ 1,
    inventory ↦  $\langle [type \mapsto \text{"MSG\_BLOCK"}, hash \mapsto \text{Cardinality}(\text{blocks})] \rangle$ 
  ]
]

MakeReject(rejected\_command)  $\triangleq$  [
  header ↦ [magic ↦ 619259748, command ↦ "reject"],
  payload ↦ [message ↦ rejected\_command]
]

MakeBlock(blocks)  $\triangleq$  [
  header ↦ [magic ↦ 619259748, command ↦ "block"],
  payload ↦ [
    version ↦ 70015,
    prev\_block ↦  $\text{Cardinality}(\text{blocks})$ ,
    merkle\_root ↦ 0,
    bits ↦ 0,
    nonce ↦ 0,
    transactions ↦  $\langle \rangle$ 
  ]
]

```
